

Database Systems (CS2005)

Date: Mon, 23 Feb 2026

Course Instructor(s)

IR, ZA, MM, MN, AA

Sessional-1 Exam

Total Time (Hrs.): 1

Total Marks: 25

Total Questions: 4

Solution

Roll No

Section

Student Signature

Instructions: Attempt all the questions on answer sheet.

Consider the following basic schema for an App Store system. The APP table maintains information about different applications available on the platform, while the APP_DEVELOPER table maintains the list of all developers.

The ACCOUNT table stores basic information for both users and developers. Each developer is associated with an account in the ACCOUNT table. The attributes DeveloperAccountID and AppID in the APP_DEVELOPER table are the foreign keys that reference in the ACCOUNT and APP table respectively. The ACTIVITY relation records the interaction between accounts and applications, including the download_date and the rating assigned to an app by different accounts. The rating is represented as a numerical value ranging from 1 to 5. Note Download_Date is NOT NULL attribute.

APP

AppID	Name	Price
101	ChatMaster	25000
102	PhotoEdit Pro	50000
103	GameZone	85000
104	FitTracker	70000
105	LearnSQL	90000
106	MusicStream	100000
107	TravelBuddy	65000

APP_Developers

AppID	DeveloperAccountID
101	2
102	3
101	5
104	7
104	2
105	3
106	5
105	7

Account

AccountID	Name	Country
1	Ali	Pakistan
2	Ahmad	Pakistan
3	Sara	USA
4	John	UK
5	Bilal	UAE
6	Ayesha	Canada
7	Noor	Australia

Activity

ActivityID	AccountID	AppID	Download_Date	Rating
1	2	104	2025-01-10	4
2	4	105	2026-01-10	1
3	6	103	2025-02-10	2
4	2	104	2025-01-10	3
5	4	105	2026-02-10	1
6	3	105	2025-01-10	5

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CLO # 4: Author queries using relational algebra and SQL

Q. No 1: Specify the following queries in SQL?

(10 points)

- a) Find the AppID of the apps that are created by two or more developers and are never downloaded.

```
SELECT AD.AppID FROM APP_Developers AD
LEFT JOIN Activity AC ON AD.AppID = AC.AppID
WHERE AC.AppID IS NULL
GROUP BY AD.AppID
HAVING COUNT(DISTINCT AD.DeveloperAccountID) >= 2;
```

- b) List the names of the Pakistani App Developers who have given a rating of 5 to their own Apps.

```
SELECT DISTINCT A.Name
FROM Account A
INNER JOIN APP_Developers AD
ON A.AccountID = AD.DeveloperAccountID
INNER JOIN Activity AC
ON AC.AppID = AD.AppID AND AC.AccountID = A.AccountID
WHERE A.Country = 'Pakistan' AND AC.Rating = 5;
```

CLO # 4: Author queries using relational algebra and SQL

Q. No 2: Consider the given database and apply the following operations. State if the operation will be carried out successfully or not. In case of successful operation, indicate the changes that will be made to the above database. Also, state all the integrity constraints violated by each operation, if any. Please note that all operations are independent.

Consider all foreign keys have on delete Cascade and on update Cascade.

(3

points)

- a. Insert into APP_Developers values(108,7)

Operation Rejected: because AppID 108 does not exist in APP table

- b. Delete from APP where name = 'FitTracker'

Operation Accepted:

Total Effected Rows: 4

APP_Developers: (Effected Rows:2 (Row 4,5)

Activity: (Effected Rows:2 (Row 1,4)

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c. Update Account Set AccountID = 8 Where AccountID = 3

Operation Accepted:

Total Effected Rows: 3

APP_Developers: (Effected Rows:2 (Row 2,6)

Activity: (Effected Rows:1 (Row 6)

Q. No 3: Consider the given above basic schema for an App Store system. **(3 points)**
Add referential integrity constraints: "On Delete Cascade" and "On update NoAction" to the foreign keys to all tables.

APP_Developers Table

ALTER TABLE APP_Developers ADD CONSTRAINT FK_AppDev_Account FOREIGN KEY (DeveloperAccountID)

REFERENCES Account(AccountID) ON DELETE CASCADE ON UPDATE NO ACTION;

ALTER TABLE APP_Developers ADD CONSTRAINT FK_AppDev_App FOREIGN KEY (AppID)

REFERENCES APP(AppID) ON DELETE CASCADE ON UPDATE NO ACTION;

Activity Table

ALTER TABLE Activity ADD CONSTRAINT FK_Activity_Account FOREIGN KEY (AccountID)

REFERENCES Account(AccountID) ON DELETE CASCADE ON UPDATE NO ACTION;

ALTER TABLE Activity ADD CONSTRAINT FK_Activity_App

FOREIGN KEY (AppID) REFERENCES APP(AppID) ON DELETE CASCADE ON UPDATE NO ACTION;

CLO # 4: Author queries using relational algebra and SQL

Q. No 4: Consider the given database and give the output in the form of DB relation. Also, explain in one sentence what the query is doing. **(9 points)**

A) select AP.AppID,Ap.Name from APP AP inner join Activity AC
on AP.AppID=AC.AppID
group by AP.AppID,Ap.Name
having count(Distinct AC.AccountID)>1 and AVG(AC.Rating)>=2

Solution: The apps that were downloaded by more than one unique user and have an average user rating of at least 2

Joining App and Activity Table

AppID	Name	AccountID	Rating
104	FitTracker	2	4
104	FitTracker	2	3
105	LearnSQL	4	1
105	LearnSQL	4	1
105	LearnSQL	3	5
103	GameZone	6	2

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GROUP BY (AppID, Name) with Aggregation

AppID	Name	Distinct Accounts	Avg Rating
104	FitTracker	1 (only account 2)	3.5
105	LearnSQL	2 (accounts 4,3)	2.33
103	GameZone	1	2.0

Apply HAVING Conditions

Final Result

AppID	Name
105	LearnSQL

B) `select Ac.AccountID from Activity AC
group by Ac.AccountID
having AVG(AC.Rating)>2
intersect
(select A.AccountID from Account A
where A.Country='Pakistan' or A.Country='USA'
union
select AD.DeveloperAccountID from APP_Developers AD
group by AD.DeveloperAccountID
having count(*)>1)`

Solution: The account IDs of users who have an average app rating greater than 2 and who are either from Pakistan/USA or are developers involved in more than one app.

Accounts with Avg Rating > 2

AccountID	Ratings
2	4, 3
4	1, 1
6	2
3	5

Result 1

AccountID
2
3

Accounts from Pakistan or USA

Result 2

AccountID
1
2
3

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Developers working on more than 1 app

DeveloperAccountID	Count
2	2
3	2
5	2
7	2

Result 3

DeveloperAccountID
2
3
5
7

Union of Result2 and Result 3 As Result 4

AccountID
1
2
3
5
7

Intersection of Result 1 and Result 4 as **Final Result**

AccountID
2
3

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C) `select A.AccountID,A.Name from (Account A
left join Activity AC on A.AccountID=AC.AccountID)
inner join APP_Developers AD on AD.DeveloperAccountID=A.AccountID
where AC.AccountID is null
group by A.AccountID,A.Name
having count(*)>1`

Solution: The developers who have never performed any activity (no downloads/ratings) but are associated with more than one app

LEFT JOIN Account and Activity

AccountID	Name	AC.AccountID
1	Ali	NULL
2	Ahmad	2
3	Sara	3
4	John	4
5	Bilal	NULL
6	Ayesha	6
7	Noor	NULL

WHERE AC.AccountID IS NULL

AccountID	Name
1	Ali
5	Bilal
7	Noor

INNER JOIN with APP_Developers and GROUP BY + HAVING COUNT(*) > 1

Final Result

AccountID	Name
5	Bilal
7	Noor